EDA: Product Revenue & Boxes Shipped Comparison

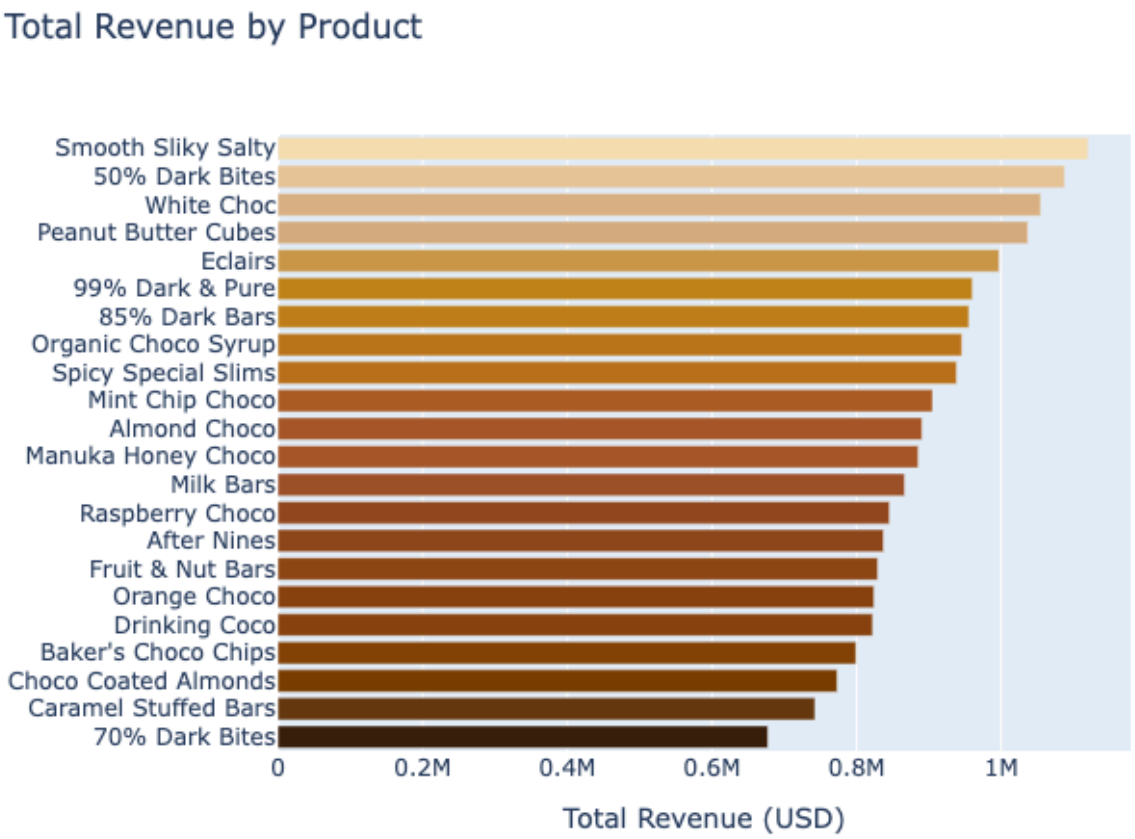
User Story 2: As a Regional Sales Director, I want to compare revenue and boxes shipped by product (e.g., 50% Dark Bites, Smooth Silky Caramel) in order to identify best-selling products and adjust inventory or promotions.

This notebook demonstrates that the Chocolate Sales dataset can support this user story by visualizing product-level revenue and shipment volume.

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
0	Jehu Rudeforth	UK	Mint Chip Choco	2022-01-04	5320.0	180
1	Van Tuxwell	India	85% Dark Bars	2022-08-01	7896.0	94
2	Gigi Bohling	India	Peanut Butter Cubes	2022-07-07	4501.0	91
3	Jan Morforth	Australia	Peanut Butter Cubes	2022-04-27	12726.0	342
4	Jehu Rudeforth	UK	Peanut Butter Cubes	2022-02-24	13685.0	184

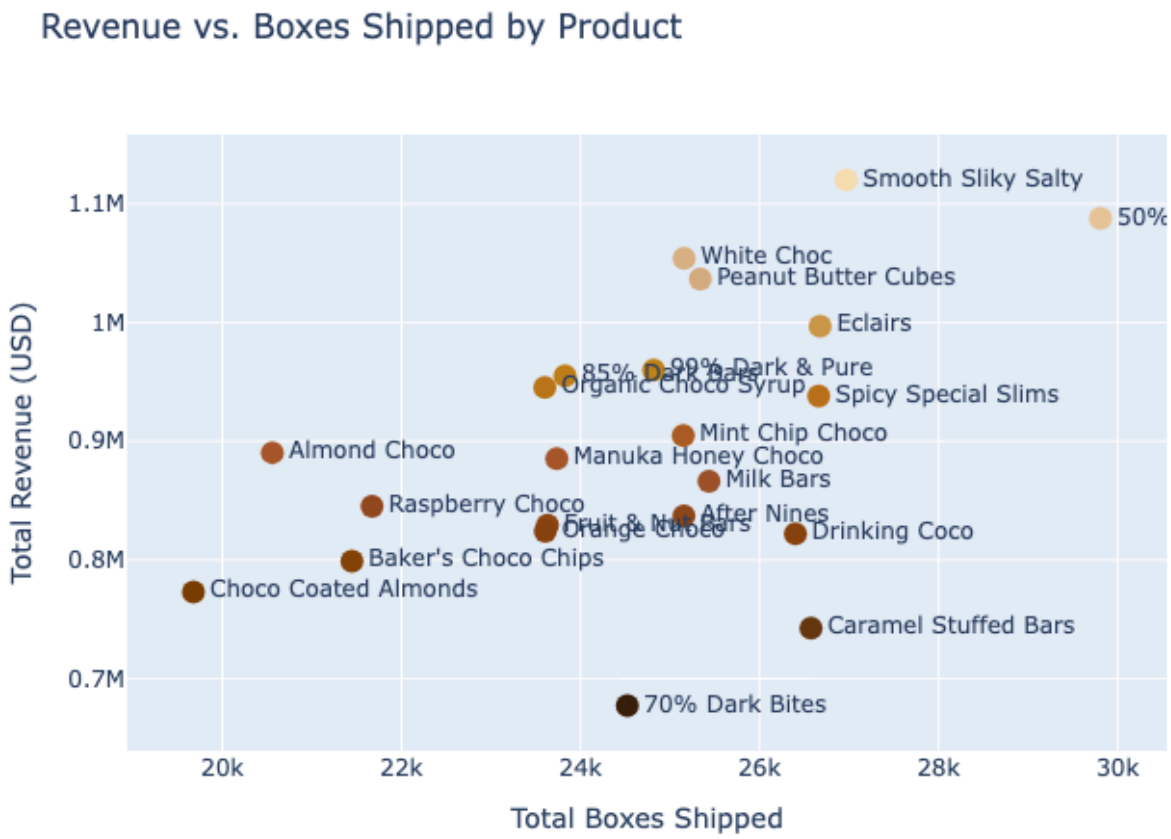
Visualization 1: Total Revenue by Product

This bar chart ranks all products by total revenue, allowing the director to quickly spot which products are the top revenue drivers and which may need promotional support or discontinuation.



Visualization 2: Revenue vs. Boxes Shipped by Product

This scatter plot compares revenue against boxes shipped for each product. Products in the upper-right are high-volume, high-revenue stars. Products with high boxes but low revenue may have pricing issues, while those with high revenue but low boxes indicate premium, high-margin items. This directly supports the director's decision on inventory allocation and promotional strategy.



Summary

These visualizations confirm the dataset supports User Story 2. The bar chart enables direct product revenue ranking, while the scatter plot reveals the relationship between shipment volume and revenue per product—both essential for a Regional Sales Director to identify best-sellers and adjust inventory or promotions accordingly. We can see a clear positive correlation between both variables, though there are some outliers